Nathan Walker, BS Electrical Engineering

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Executive Summary

Systems-focused Electrical Engineer with experience in structured testing, verification, data analysis, and cross-functional engineering collaboration. Skilled in developing and executing test procedures, ensuring data integrity, and supporting system validation from development through evaluation. Experienced working with hardware and software systems in research environments requiring accuracy, traceability, and disciplined execution of experimental workflows.

Work Experience

Graduate Researcher May 2024 – Present

Thesis: *Design and Implementation of Adaptive Active Snore Reduction Using Near-Field Array Processing*

- Developed and maintained structured experimental workflows to ensure accurate, repeatable data collection and system validation.
- Executed verification testing and analyzed experimental data using MATLAB and C to support performance evaluation and decision-making.
- Identified, diagnosed, and resolved undocumented hardware and system issues through systematic troubleshooting and root-cause analysis.
- Supported data integrity and reliability through careful experimental setup, documentation, and controlled testing procedures.
- Communicated technical findings and experimental results through formal reviews and cross-functional collaboration.

ASPIRE Research Assistant May 2023 – April 2024

- Supported structured hardware testing and validation of PCB assemblies and electronic subsystems using laboratory instrumentation.
- Developed and executed test procedures to ensure consistent evaluation of system functionality and performance.
- Assembled and integrated electronic hardware while maintaining adherence to design requirements and experimental constraints.
- Created technical documentation supporting repeatable testing, system traceability, and experimental reproducibility.
- Coordinated multiple concurrent projects, tracking progress and ensuring completion of experimental milestones.

Engineering Projects

International Future Energy Challenge – Team Lead July 2023 – July 2024

- Translated system-level objectives into testable hardware requirements and coordinated subsystem development across team members.
- Performed hardware integration and validation testing to ensure system performance met defined design goals.
- Conducted structured troubleshooting and root-cause analysis to resolve system-level integration issues.
- Documented engineering decisions and presented verification results during formal design reviews and project evaluations.

Skills

Altium/KiCad	C/C++ & Python	MATLAB/Data Analysis
Linux/GitHub/Scripting	Solid Edge/ Mechanical Design	Hardware Troubleshooting

Education

Utah State University – Logan Utah

- M.S. Electrical Engineering – 3.83 GPA May 2026
- B.S. Electrical Engineering – 3.97 GPA May 2025